

LAMASY

ERP MODERN UNTUK LAUNDRY UMKM INDONESIA

Dokumen Internal — Operasional & Teknis

Untuk tim teknis, ops, support, finance

Versi: Production — lamasy.harpy.id

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Klasifikasi: INTERNAL — Confidential

Daftar Isi

1. Pengantar

LAMASY adalah platform SaaS multi-tenant untuk operasional laundry, dirancang dari UMKM single-outlet sampai chain multi-outlet. Sistem dibagi tiga lapisan akses: Outlet (operasional harian), HQ (manajemen multi-outlet owner), dan SuperAdmin (admin platform LAMASY).

Dokumen ini ditujukan untuk tim internal — engineering, ops, finance, dan support. Berisi detail teknis, flow operasional, dan referensi sistem yang tidak akan dibagikan ke pihak eksternal.

2. Arsitektur Global

2.1 Model Multi-Tenant

Sistem menggunakan model shared database + shared schema dengan scoping per tabel via kolom tenant_id (dan outlet_id untuk transaksional). Setiap tenant adalah satu bisnis laundry dengan satu owner dan satu atau lebih outlet. Outlet adalah lokasi fisik dengan operasional independen.

- **Tenant** = bisnis laundry (1 owner, N outlet)
- **Outlet** = lokasi fisik dengan operasional independen
- **Isolasi data** enforced di query layer via TenantResolver + TenantQuery helper
- **~125 tabel database** mencakup operasional, finansial, HR, AI, dan platform

2.2 Tech Stack

Layer	Tech	Catatan
Backend	PHP 8 vanilla	No framework, simple deployment
Database	MariaDB Hostinger	Shared hosting, single instance
Frontend	Vanilla HTML/CSS/JS + Ponytail design system	Tema dark navy + teal
Auth	Session + HMAC-SHA256 + CSRF	Per-request token
AI	Anthropic Claude	claude-sonnet-4-x, claude-haiku-4-5
Email	Hostinger SMTP via PHPMailer	noreply@harpy.id
WhatsApp	Fonnte API	1000 msg/bulan free tier
PWA	manifest + service worker	Installable, offline-first cache
Hosting	Auto-deploy via git push	~15 detik deploy time
Timezone	Asia/Jakarta (UTC+7)	Critical untuk laporan harian

2.3 Folder Structure

/	– Outlet level pages (root)	/core/	–
Reusable libraries (30 file)	/assets/banners/	– Banner image uploads	
(SA) /api/	– Internal AJAX endpoints	/hq/	–

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HQ-level pages (30 file)  /superadmin/          - SuperAdmin pages (24 file)
/middleware/              - Auth/permission guards  /db/              - Schema +
migrations  /docs/        - Documentation  /master/          -
Hostinger config (gitignored)
```

2.4 Permission Model — 3 Lapisan

- **SuperAdmin** — Founder + tim LAMASY. RBAC dengan 5 default role + 29 permissions.
- **HQ** — Owner bisnis. Akses via hq.access permission.
- **Outlet** — Kasir, kurir, staff. Role-based (owner, manager, kasir, kurir, produksi).

3. Flow Operasional

Section ini menggambarkan 12 flow utama yang berjalan di sistem. Setiap flow include actor, langkah, dan data yang berubah.

3.1 Tenant Onboarding Flow

Dari calon customer datang sampai jadi tenant aktif yang transaksi:

1. Visitor buka lamasy.harpy.id → klik 'Daftar Gratis'
2. Isi form /register.php (nama bisnis, owner, email, WA, password, nama outlet pertama)
3. DB INSERT saas_tenants + hl_outlets + saas_users; auto-credit 10.000 coin trial
4. SaNotifier::tenantRegistered → email ke SA dengan permission registrations.view
5. Email verification link dikirim → tenant klik → status = trial
6. Login pertama kali → /accept-tos.php → Splash tips + Product Tour walkthrough
7. TenantProvisioner seed default data (layanan, parfum, bahan, COA)
8. Transaksi pertama via POS → status = aktif
9. Setelah 7 hari trial: top-up coin (tetap aktif) atau habis (grace 3 hari → suspended)

3.2 POS / Order Lifecycle

Order dari penerimaan sampai customer pulang dengan cucian bersih:

10. Customer datang → kasir buka /pos.php
11. Pilih customer existing atau input no.HP baru + parfum (opsional)
12. Add layanan (cuci complete, setrika, dll). Bisa inline-create kalau belum ada.
13. Tier express: Regular / Express (+harga). Antar checkbox toggle ke kurir flow.
14. Estimasi otomatis via applyMaxEstimasi() — max dari semua layanan + tier override
15. Auto-apply promo aktif, cek saldo deposit + member tier discount
16. Pembayaran: cash / transfer / deposit-deduct
17. DB INSERT hl_order + hl_order_item + hl_pembayaran; status = 'diterima'
18. Auto-deduct stok bahan (FinancialCalculator); auto-issue poin loyalty
19. Cetak struk via StrukGenerator.php + QR code untuk track + portal
20. Staff produksi scan QR di /produksi.php → 6 tahap dengan signature + foto
21. Tahap: Terima → Sortir → Cuci → Setrika → Packing → Selesai
22. Status order auto-update: diterima → diproses → siap
23. Auto-send WA 'Cucian siap diambil'
24. Customer ambil sendiri (mark 'selesai') atau via antar-jemput (flow 3.3)

3.3 Antar-Jemput Flow

Order dengan pickup/delivery oleh kurir outlet:

25. POS dengan checkbox Antar=ON → auto-create hl_antar_jemput
26. Type: pickup, delivery, atau both. Zona auto-detect dari alamat customer.
27. Dispatcher buka /antar-jemput.php → assign kurir aktif
28. Kurir terima notif WA 'Task baru: pickup dari Customer X'
29. Kurir buka /kurir mobile → step buttons: Berangkat → Sampai → Pickup done
30. Kembali outlet → produksi selesai → status order = siap
31. Auto-create antar-jemput delivery (kalau type=both)
32. Kurir delivery: step buttons + Done modal (foto bukti + customer signature)
33. Status order = selesai, loyalty poin issued

3.4 Self-Service Mesin Flow

34. Customer datang ke outlet laundromat mode

- 35. Scan QR mesin → /self.php?m=KODE (public, no login)
- 36. Form: input HP + pilih cycle (cuci/keringkan/keduanya)
- 37. Submit → DB transaction race-safe: SELECT FOR UPDATE → INSERT sesi → UPDATE mesin
- 38. Customer mulai cycle → timer countdown
- 39. Cycle selesai → status = 'selesai', alert di dashboard staff
- 40. Staff verify + reset → status = 'tersedia'

3.5 Customer Portal Flow

- 41. Customer scan QR di struk → /p?t=<portal_token>
- 42. Token validasi via hl_pelanggan.portal_token (UNIQUE, auto-gen)
- 43. Set session pelanggan_id → redirect /pelanggan (read-only)
- 44. Tab: Order Aktif / History / Hadiah / Saldo Deposit
- 45. Member tier badge + poin balance + reward catalog
- 46. Klik order detail → timeline status + item breakdown + estimasi
- 47. Owner bisa regenerate token kalau leaked → token lama invalid

Portal customer READ-ONLY by design — tidak bisa edit/order untuk keamanan.

3.6 Coin / Billing Flow

- 48. Tenant baru: auto-credit 10.000 coin (trial gratis 7 hari)
- 49. Tenant transaksi normal: fitur basic gratis
- 50. Tenant pakai fitur AI / premium: coin deducted per call (saas_coin_pricing)
- 51. AIRateLimiter cek limit harian per fitur sebelum allow call
- 52. Saldo coin < threshold (5.000): notif WA + email + banner dashboard tenant + alert SA
- 53. Tenant top-up: buka /deposit → pilih package → upload bukti transfer
- 54. SA Finance role approve di /superadmin/clients.php → UPDATE coin_balance + INSERT coin_ledger
- 55. SaNotifier::paymentReceived → confirmation email ke tenant
- 56. Kalau coin habis tanpa top-up: enter grace 3 hari → SaNotifier::trialExpiring
- 57. Grace habis → status = suspended → block semua page kecuali billing + support
- 58. Top-up → reactivate

3.7 AI Request Flow

- 59. User trigger AI feature (mis. HQ owner klik 'Generate Briefing')
- 60. AIRateLimiter::checkAndIncrement → cek hl_ai_usage vs saas_coin_pricing.daily_limit
- 61. Over limit → return error + block. OK → INSERT/UPDATE counter.
- 62. Cache check (hl_ai_cache + Anthropic prompt cache 5-min TTL)
- 63. Cache hit → return cached. Miss → call Claude API.
- 64. AnthropicClient::send dengan AIPersona system prompt + data injection
- 65. Response: log usage (tokens_in/out, cost_idr, cache_hit) + deduct coin
- 66. INSERT hl_ai_usage + coin_ledger entry → return ke UI
- 67. SA tracking di /superadmin/ai_usage.php: cost vs revenue, margin per fitur/tenant

3.8 Smartlink Migration Flow

Tenant existing di kompetitor Smartlink pindah ke LAMASY dengan data Excel:

- 68. Tenant request migration ke SA
- 69. SA buka /superadmin/migrations.php → 'New Migration Job'
- 70. Upload file Excel hasil export Smartlink (2-tier header, multi-item nota)

71. MigrationImporter::parseExcel → extract kolom + rows
72. AI mapping via AIMigrationMapper (Claude analyze kolom Excel vs schema LAMASY)
73. Cache template di hl_migration_mapping_templates untuk reuse
74. SA review mapping → confirm/edit field assignment
75. Dry run 10 rows pertama → cek correctness
76. Confirm → batch INSERT hl_pelanggan, hl_order, hl_order_item
77. Hitung kolom 'Total' (heuristic — handle biaya tambahan)
78. Job logged di hl_migration_jobs dengan status success/partial

Migration upload tidak deduct coin — gratis service untuk acquisition.

3.9 Support Ticket Flow

79. Tenant ada masalah → HQ owner buka /hq/support → 'New Ticket'
80. Subject, body, attachment opsional → submit
81. SaNotifier::supportTicketCreated → email ke SA dengan permission support.view
82. SA Support staff buka /superadmin/support.php → inbox semua tickets
83. Filter: open / replied / resolved → klik ticket → detail thread
84. Reply (perm support.reply): tulis response → auto-send notif email + WA
85. Multiple exchanges (thread)
86. Mark closed (perm support.close): status = resolved → tenant bisa rate response

3.10 RBAC SuperAdmin Flow

87. Founder hire staff baru (mis. Support agent)
88. Owner SA buka /superadmin/settings.php → tab 'Roles & Permissions'
89. Pakai default 'support' role (6 perms) atau create custom
90. Buka tab 'SA Team' → '+ Tambah' → input data + assign role
91. DB INSERT super_admins + assign role_id
92. Staff baru login → SaPermission::loadIntoSession() load perms dari junction table
93. Setiap page SA: SaPermission::require('feature.action')
94. Notif routing: SaNotifier::resolveRecipients(\$event) match permission
95. Filter by notify_enabled + notif_events opt-in → send email (throttled 60s)

3.11 Multi-Outlet HQ Impersonate Flow

96. Owner buka /hq/dashboard → lihat 'Kesehatan Outlet' card
97. Klik 'Masuk' per outlet → link include /switch-outlet?outlet=N&t=<HMAC-token>
98. Token = hash_hmac('sha256', \$outlet_id, \$secret), validity ~60s
99. Server verify token → set \$_SESSION['outlet_id'] = N
100. Redirect /dashboard outlet view → tenant_guard cek session
101. Owner do anything outlet-level
102. Switch ke HQ View → clear outlet_id session

3.12 Banner Carousel Flow

103. SA buka /superadmin/banners.php → '+ Tambah Banner'
104. Pilih mode: Gambar (upload 1600×400 optimal, max 800 KB) atau Gradient
105. Set: judul, deskripsi, CTA label+URL, icon emoji, schedule, urutan
106. Save → INSERT saas_banners (image_url OR bg_gradient)
107. Tenant buka /dashboard → BannerLoader::activeForTenant query banner aktif
108. WHERE is_active=1 AND between starts_at/ends_at, LIMIT 5
109. Render carousel: image mode bg-image + dark overlay; gradient CSS bg

110. Auto-rotate setiap 6 detik + dots indicator

4. Layer 1: Outlet (Operasional Harian)

Audience: Kasir, kurir, staff produksi, manager outlet, owner. ~25 page di root project.

4.1 Modul Transaksi & Order

Page	Fungsi
pos.php	POS kasir — input order baru
orders.php	Daftar order + filter status
kanban.php	Kanban view drag-drop status
produksi.php	Staff produksi — 6 stage workflow
mesin.php	Master mesin + monitor real-time

4.2 Antar-Jemput

Page	Fungsi
antar-jemput.php	Dispatcher view
kurir.php	Kurir mobile view (step-button + done modal)
kurir-master.php	CRUD kurir + account

4.3 Inventori & Master

Page	Fungsi
inventori.php	Stok bahan + mutasi + PO PDF
layanan.php	Master layanan + kategori + tier
promo.php	Promo dengan apply outlet
customer.php	Master customer
member.php	Member tier system
deposit.php	Wallet customer
loyalty.php	Sistem poin + reward
retention.php	Dormant customer

4.4 Customer-Facing Public

Page	Fungsi
pelanggan.php	Portal home
pelanggan-order.php	Order detail
self.php	Self-service mesin booking
track.php / cek.php	Public order tracking

p.php	Portal token entry
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4.5 Kas, Keuangan & HR

Page	Fungsi
kas.php	Kas harian + mutasi
laporan.php	Laporan SAK EMKM (Neraca/LR/Arus Kas)
piutang.php	Piutang B2B
karyawan.php	Master karyawan
absensi.php	Clock-in/out

4.6 Role di Outlet

Role	Akses
owner	Full access semua fitur outlet
manager	Operasional + sebagian finansial
kasir	POS + customer + payment
kurir	Mobile view + task list (login redirect ke /kurir)
produksi	Produksi page only

5. Layer 2: HQ (Multi-Outlet Owner)

Audience: Owner bisnis. Akses via `hq.access` permission. ~30 page di `/hq/`.

5.1 Dashboard & Konsolidasi

Page	Fungsi
<code>dashboard.php</code>	Konsolidasi metrik semua outlet, kesehatan outlet, switch impersonate
<code>outlet.php</code>	CRUD outlet, setting antar-jemput, zona kurir

5.2 Finansial Konsolidasi

Page	Fungsi
<code>keuangan.php</code>	Laporan SAK EMKM gabungan
<code>laporan.php</code>	Custom report konsolidasi
<code>mutasi.php</code>	Transfer kas antar outlet
<code>billing.php</code>	Konsolidasi billing
<code>coin-info.php</code>	Info coin platform (saldo, history, top-up)

5.3 Operasional Lintas Outlet

Page	Fungsi
<code>inventori.php</code>	Stok konsolidasi + transfer antar outlet
<code>mesin.php</code>	Mesin konsolidasi + report utilization
<code>karyawan.php</code>	Semua karyawan lintas outlet
<code>penggajian.php</code>	HQ-level payroll
<code>pelanggan.php</code>	Customer lintas outlet
<code>droppoint.php</code>	Drop point manager

5.4 AI Tools (HQ-only)

Page	Fungsi
<code>ai-chat.php</code>	Owner tanya-jawab data natural language
<code>ai-churning.php</code>	Churn risk detection

5.5 HQ vs Outlet — Perbedaan

Aspect	Outlet	HQ
Scope data	1 outlet	semua outlet tenant
Karyawan view	yang assigned	semua karyawan

Finansial	per-outlet	konsolidasi
Master data	inherited dari HQ	source of truth
AI tools	briefing outlet	chat full + churn
Switch outlet	tidak bisa	bisa impersonate

6. Layer 3: SuperAdmin (Platform Admin)

Audience: Founder + tim LAMASY (ops/finance/support/viewer). ~24 page di /superadmin/.

6.1 Modul Utama

Kategori	Pages	Fungsi
Dashboard & Health	dashboard, health	Total tenant, revenue, churn, AI abuse alerts
Client Management	clients, client_detail, impersonate	Tenant + outlet CRUD, suspend/topup
Onboarding	registrations, onboarding, migrations	Pending queue + manual + Smartlink import
Finance	billing, payments, coin_pricing, packages	Revenue tracking + pricing config
AI	ai_usage	Cost vs revenue margin per fitur/tenant
Communication	support, announcements, broadcast, banners	Tickets + mass comms + banners
Settings	settings (multi-tab)	Maintenance, ToS, Tips, Notif, SA Team, Roles

6.2 RBAC SuperAdmin — 5 Default Role

Role	Perm Count	Untuk
owner	29 (all)	Founder Rizky
ops	7	Registrasi + onboarding + impersonate
finance	10	Billing + payments + pricing + packages
support	6	Tickets + churn + broadcast
viewer	10	Dashboard + metric read-only

6.3 29 Permissions (Sample)

- super_admins.manage — kelola SA team
- clients.view / clients.suspend / clients.topup
- support.view / support.reply / support.close
- billing.view / billing.refund
- coin_pricing.edit, packages.manage
- announcements.publish, banners.publish, broadcast.send
- registrations.view / registrations.approve, migrations.run

7. Integrasi AI

7.1 9 Fitur AI yang Live

Fitur	Modul	Layer	Pricing
AI Briefing harian	dashboard widget	Outlet + HQ	varies
AI Chat	conversational data query	HQ	per-call
AI Churning	churn risk detection	HQ	per-call
AI Insight	pattern + anomaly	HQ	per-call
AI Migration Mapper	Excel column matching	SA only	free 1x
AI Anomaly Alerts	real-time transaction watch	Outlet	per-call
AI Daily Report	auto-generated summary	HQ	per-call
AI Generate Nota	smart nota template	Outlet	per-call
AI Send WA	AI-rewritten WA message	Outlet	per-call

7.2 Anti-Abuse: AIRateLimiter

core/AIRateLimiter.php cek 3 hal sebelum allow call:

- DB-backed counter (hl_ai_usage per tenant + feature + date)
- Daily limit cek (saas_coin_pricing.daily_limit per fitur)
- Cost ceiling cek (cost yang akan terjadi vs sisa coin balance)

Tenant yang hit limit 3+ kali sehari → flagged di SA dashboard dengan violet AI shimmer banner.

7.3 Tracking & Margin

/superadmin/ai_usage.php menampilkan:

- Total AI calls + cache hit rate
- Total tokens in/out
- Cost Anthropic (actual API cost dalam IDR)
- Revenue Coin (coin spent × harga per coin)
- Net Margin per fitur + per tenant
- Trend harian (line chart cost vs revenue)

8. Database Schema Overview

103 tabel total, dikelompokkan berdasarkan domain.

8.1 SaaS Platform Tables (saas_* / super_*)

- saas_tenants — tenant master + coin balance
- saas_users — user accounts per tenant
- saas_banners — banner carousel (gradient ATAU image)
- saas_coin_pricing — coin pack + AI limits
- saas_sa_notif_log — SA notification audit
- saas_packages, saas_pricing — subscription tiers
- super_admins — SA accounts dengan role_id
- sa_roles, sa_permissions, sa_role_permissions — RBAC junction

8.2 Tenant Operational (hl_*)

Semua tabel hl_* scoped tenant_id (+ outlet_id untuk transaksional).

Master data

- hl_outlets, hl_layanan, hl_layanan_master, hl_parfum
- hl_bahan, hl_bahan_mutasi, hl_bahan_stok (view)
- hl_mesin, hl_mesin_cycle, hl_mesin_sesi
- hl_pelanggan, hl_pelanggan_member, hl_member_tier
- hl_drop_points, hl_kurir, hl_karyawan_outlet

Transaksional

- hl_order, hl_order_item, hl_order_notes
- hl_pembayaran
- hl_proses_input — produksi 6-stage audit
- hl_antar_jemput — pickup/delivery

Finansial

- hl_kas, hl_kas_bank, hl_kas_bank_mutasi
- hl_coa, hl_jurnal_manual
- hl_aset_tetap, hl_liabilitas, hl_piutang
- hl_laporan_cache

HR & Payroll

- hl_absensi, hl_izin
- hl_gaji, hl_gaji_komponen, hl_komisi_rekap
- hl_bonus_rule, hl_bonus_rule_outlet

Loyalty & Promo

- hl_promo, hl_promo_outlets
- hl_loyalty_log, hl_poin_reward, hl_poin_reward_outlet
- hl_deposit_topup, hl_deposit_usage, hl_deposit_refund, hl_deposit_bonus_tier
- hl_express_tier

AI Tracking

- hl_ai_usage — per-tenant per-feature daily counter
- hl_ai_cache — response cache

- hl_ai_outreach_log — AI-generated outreach

Lainnya

- hl_audit_log — semua action sensitif
- hl_login_attempts, hl_notif_log
- hl_broadcast, hl_broadcast_recipient
- hl_checklist_template, hl_checklist_submission
- hl_migration_jobs, hl_migration_mapping_templates
- hl_delete_request — UU PDP deletion requests
- hl_permissions, hl_splash_seen, hl_splash_tips
- coin_ledger, email_verifications

9. Security & Compliance

9.1 Authentication

- Session-based dengan PHP native sessions
- Password hashed dengan bcrypt via `password_hash()`
- Login attempts logged di `hl_login_attempts` (brute-force protection)
- Email verification wajib sebelum aktif

9.2 Authorization

- CSRF token per request (HMAC-SHA256 signed)
- Tenant scoping enforced di `TenantQuery::scoped()`
- Permission check di setiap action via `hl_permissions` / `SaPermission`
- HQ guard via `hq.access` permission

9.3 Multi-Tenant Isolation

- Setiap query include `WHERE tenant_id = $current_tenant`
- Cross-tenant data access prevented at query layer (periodic audit)
- Outlet scoping pada transaksional tables

9.4 Compliance Indonesia

- **UU PDP No 27/2022** — Personal Data Protection (delete request mechanism)
- **UU PK 8/1999** — Consumer Protection
- **UU ITE 11/2008** — Electronic Information & Transactions
- **SAK EMKM** — laporan keuangan auto-generate compliant

9.5 Audit Trail

- `hl_audit_log` — log semua action sensitif
- `super_admin_audit` — SA-level action log
- Authentication events, permission changes, financial transactions

10. Komunikasi & Notifikasi

10.1 Channels

Channel	Provider	Use Case	Limit
Email	Hostinger SMTP	Verification, password reset, SA notif, broadcast	—
WhatsApp	Fonnte API	Order siap, kurir assignment, broadcast, low coin	1000 msg/bulan free
In-App	Native	Splash tips, banner carousel, Product Tour, alerts	—

10.2 SA Notif Routing

core/SaNotifier.php route notif berdasarkan permission match:

- 6 event types: tenantRegistered, emailVerified, outletActivated, supportTicketCreated, trialExpiring, outletSuspended
- Filter recipient via SaPermission match per event type
- Per-SA opt-in via super_admins.notify_enabled + notif_events CSV
- Throttle 60-detik dedup per event_type untuk avoid spam

11. Operasional Tim

11.1 Deploy Workflow

111. Edit code lokal
112. Test via MCP browser (Chrome extension) untuk verifikasi visual
113. git commit + git push origin main
114. Auto-deploy ke production ~15 detik
115. Verify di production via MCP browser smoke test

11.2 DB Access untuk Tim

- Production DB via mysql client + ~/.my.cnf (kredensial Hostinger)
- Path: /opt/homebrew/opt/mysql-client/bin/mysql
- Backup: Hostinger automated daily backup

11.3 Master Credentials

- master/config/db.php — gitignored, never commit
- SuperAdmin Rizky — owner role, full access
- Fonnte API key + Anthropic API key di env config

11.4 Monitoring & Health

- /superadmin/health.php — real-time platform health
- WA delivery rate alert (di bawah 95% trigger banner)
- Error log via core/ErrorLogger.php
- AI usage abuse detection

11.5 Common Operasional Tasks

Task	Lokasi
Onboard tenant baru	/superadmin/registrations.php
Top up coin tenant	/superadmin/clients.php → Topup
Suspend tenant	/superadmin/clients.php → Suspend
Migrasi Excel Smartlink	/superadmin/migrations.php
Reply support ticket	/superadmin/support.php
Broadcast pengumuman	/superadmin/announcements.php
Update banner dashboard	/superadmin/banners.php
Add SA team member	/superadmin/settings.php → SA Team
Toggle maintenance mode	/superadmin/settings.php → Maintenance
Cek AI usage + margin	/superadmin/ai_usage.php

12. Roadmap Internal

12.1 High Priority

- **2FA TOTP** untuk SuperAdmin accounts — security hardening
- **Audit Log Viewer** terdedicated untuk SA — searchable + exportable
- **Extend Trial Manual + Manual Coin Adjustment** UI di SA — operational efficiency
- **Email Template Editor** — jangan hardcode template di Mailer.php

12.2 On Hold

- **Native app** (Capacitor thin shell + remote webview) — hold sampai PWA feature-complete
- **QRIS + WhatsApp Business API** — hold separately, evaluasi business case dulu

12.3 Exploratory

- Lighthouse baseline metrics untuk measure improvement
- Cloudflare Analytics integration
- CF panel optimization: Auto-Minify + Brotli + Tiered Cache (manual setting)

13. Design System

13.1 Outlet + HQ — Ponytail

Token	Value	Usage
Background	Dark navy #0F1C3A	Page background
Brand Accent	Teal #35E8D5	Primary CTA, active state, brand
Type	Plus Jakarta Sans + DM Mono	Body + tabular
Vibe	Operational ERP	Mobile-first, PWA-ready

13.2 SuperAdmin — AI Command Center

Token	Value	Usage
Background	Obsidian #0A0F1F + radial gradient + dot-grid	Page bg
Surface	Slate #141B2D dengan glass blur	Cards
Brand Accent	Teal #35E8D5	CTA, active state
AI Accent	Violet #A78BFA	AI-specific features semantic
Type	Inter + Inter Tight + JetBrains Mono	Body + display + data
Signature	AI Shimmer (teal→violet animated)	Borders, pills, strips
Vibe	Premium AI tooling	Modern dashboard

14. Stats Sistem

Metric	Count
DB Tables	~125
PHP files (Outlet + Public)	61
PHP files (HQ)	30
PHP files (SuperAdmin)	24
Core Libraries	30
AI Features Integrated	9
Default SA Roles	5
SA Permissions	29
Auto-deploy Time	~15 detik

15. Update Fitur — Q2 2026

Modul & penyempurnaan yang live setelah 24 Juni 2026. Semua opt-in / aman dan terintegrasi ke modul terkait. Detail spec & plan ada di docs/superpowers/.

15.1 Referral / Ajak Teman

- Opt-in per owner (tenants.referral_enabled + butuh loyalty aktif). Diatur di HQ → Loyalty.
- Pelanggan punya kode referral (hl_pelanggan.referral_code) + link share ?ref= ke self-booking.
- Atribusi 2 jalur: input kode manual di POS/self-booking ATAU link share. Tabel hl_referral (UNIQUE referee = satu teman sekali).
- Payout: pengajak + teman dapat poin saat order pertama teman LUNAS (idempoten, via Loyalty::adjust). Cap per-pengajak; cap penuh → teman tetap dapat, pengajak 0.
- Anti-abuse: teman harus baru (tak punya order), no self-refer (id+telepon). core/Referral.php.

15.2 Voice Order (POS)

- Kasir input order via suara: rekam → STT (native Android) → api/voice_order_parse.php → core/VoiceOrderParser.php (Claude) → form POS terisi.
- Tombol mic berubah state saat recording; potong coin hanya saat parse sukses. Termasuk dalam metering AI.

15.3 Offline POS + Sync

- POS tetap menerima order saat internet mati. assets/offline-pos.js: IndexedDB (katalog cache + antrian order).
- Endpoint pos.php?action=catalog_snapshot & sync_offline; core/OrderCreator.php::createOffline. Idempoten via hl_transaksi.offline_uuid (UNIQUE).
- Struk offline pakai kode sementara OFF-<dev>-<seq>; saat sync dapat nomor asli + alias offline_ref (tetap bisa dilacak di track.php/orders.php).
- Service worker sw-tenant.js cache shell POS. Kontrol online-only (deposit/redeem/voucher) auto-disable saat offline; network-fail saat online → fallback enqueue.

15.4 Checklist Wajib-Foto

- Saat buat template checklist, tiap item bisa ditandai 'wajib foto'. core/Checklist.php; data di items_json/answers_json (tanpa migrasi).
- Sisi pengisian: item wajib-foto yang dicentang harus lampirkan foto — blok simpan (validasi klien + server otoritatif). Upload ke uploads/foto_checklist (anti-XSS path).

15.5 Absensi: Selfie + Geofence + Shift

- Clock-in: selfie WAJIB + geofence ketat (config per-outlet + peta Leaflet). Lokasi diverifikasi server.
- Jadwal shift (hl_shift / hl_jadwal_shift): hitung telat/lembur otomatis (core/ShiftCalc.php), snapshot saat absen, handover shift.

15.6 Penggajian: Bonus & Bagi-Hasil

- Aturan bonus karyawan (hl_bonus_rule, core/BonusEvaluator.php) masuk ke slip gaji.
- Bagi-hasil (core/BagiHasilCalculator.php, hl_bagi_hasil) untuk skema profit-sharing.

15.7 Ringkasan Fitur Baru

Fitur	Inti	Status
Referral	Kode/link → poin dua-duanya saat order pertama teman lunas (opt-in)	Live
Voice Order	Suara → AI parse → form POS (metered coin)	Live
Offline POS	Order saat internet mati → IndexedDB queue → sync idempoten	Live
Checklist Wajib-Foto	Item wajib lampirkan foto, blok simpan kalau kosong	Live
Absensi Hardening	Selfie + geofence + jadwal shift (telat/lembur)	Live
Bonus & Bagi-Hasil	Aturan bonus + profit-sharing ke slip gaji	Live

E2E manual di device masih pending untuk: offline POS (airplane mode), voice order, checklist wajib-foto, referral.

— End of Internal Document —